Learning Stable Constitutive Model Through Energetic Formulation

Kaushik Bhattacharya[1] Mayank Raj[2] Andrew Stuart[2]

[1]Mechanical & Civil Engineering, California Institute of Technology
[2]Computing & Mathematical Sciences, California Institute of Technology

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Context

- Constitutive Model:

  "Response as a function of load"

- Conservation Law: Equilibrium condition

Conservation Law	&	**Constitutive Model**	$\Longrightarrow$	**2nd Order PDE**
$\nabla \cdot \sigma = 0$	&	$\sigma = \nabla u$	$\Longrightarrow$	$\nabla^2 u = 0$

- Operator Learning: Learning the PDE
- Alternatively, learn the constitutive model, given the balance law.

Outline

- Problem Description
- Synthetic Data Generation
- Learning Methodology
- Visual Results
- Symmetry in the Learning Problem

Problem Setting

$$\sigma^{\dagger}(t) = \Psi^{\dagger}(\{\varepsilon(s)\}_{s=0}^{t}); \qquad \text{(Non-markovian, History dependent)}$$

$$\sigma^{\dagger}(t) = f^{\dagger}(\varepsilon(t), \xi^{\dagger}(t)), \qquad \text{(Markovian)}$$
$$h^{\dagger}(\dot{\xi^{\dagger}}(t)) = g^{\dagger}(\varepsilon(t), \xi^{\dagger}(t)), \quad \xi^{\dagger}(0) = 0;$$

$$\sigma^{\dagger}(t) = \frac{\partial \mathcal{W}^{\dagger}(\varepsilon(t), \xi^{\dagger}(t))}{\partial \varepsilon}; \qquad \text{(Energetic Formulation)}$$
$$\frac{\partial \mathcal{D}^{\dagger}(\dot{\xi^{\dagger}}(t))}{\partial \dot{\xi^{\dagger}}} = -\frac{\partial \mathcal{W}^{\dagger}(\varepsilon(t), \xi^{\dagger}(t))}{\partial \xi^{\dagger}}, \; \xi^{\dagger}(0) = 0;$$

Learning Objective

$$\sigma^\dagger(t) = \frac{\partial \mathcal{W}^\dagger(\varepsilon(t), \xi^\dagger(t))}{\partial \varepsilon} \tag{1a}$$

$$\frac{\partial \mathcal{D}^\dagger(\dot{\xi}^\dagger(t))}{\partial \dot{\xi}^\dagger} = -\frac{\partial \mathcal{W}^\dagger(\varepsilon(t), \xi^\dagger(t))}{\partial \xi^\dagger}, \quad \xi^\dagger(0) = 0. \tag{1b}$$

Objective: Learn approximations of $\mathcal{W}^\dagger$ and $\mathcal{D}^\dagger$ from a dataset.

$$\sigma = \frac{\mathcal{W}(\varepsilon, \xi; \theta)}{\partial \varepsilon}$$

$$\frac{\partial \mathcal{D}(\dot{\xi}; \alpha)}{\partial \dot{\xi}} = -\frac{\partial \mathcal{W}(\varepsilon, \xi; \theta)}{\partial \xi}, \quad \xi(0) = 0.$$

Dataset

$\{(\varepsilon_i, \sigma_i^\dagger)\}_{i=1}^N$; (Trajectories)

$\varepsilon : [0, T] \to \mathbb{R}^d$; $\sigma^\dagger : [0, T] \to \mathbb{R}^d$

Dataset & Physical Interpretation

Sources of the Dataset $\{(\varepsilon_i, \sigma_i^\dagger)\}_{i=1}^N$

- Two-scale homogenization
- Lab experiments
- **Synthetic dataset**

$$\sigma = \frac{\mathcal{W}(\varepsilon, \xi)}{\partial \varepsilon}$$
$$\frac{\partial \mathcal{D}(\dot{\xi})}{\partial \dot{\xi}} = -\frac{\partial \mathcal{W}(\varepsilon, \xi)}{\partial \xi}, \quad \xi(0) = 0.$$

Desired properties of potentials $\mathcal{W}$ and $\mathcal{D}$

- $\mathcal{W} : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is convex.
- $\mathcal{D} : \mathbb{R}^\xi \to \mathbb{R}$ is convex.

Remark: Input Convex Neural Network (Amos et al. arXiv:1609.07152)

Data Generation

Synthetic Dataset Generation

- Generate samples of $\varepsilon_i : [0, T] \to \mathbb{R}^d$ using stochastic process

$$\sigma^\dagger = \frac{\mathcal{W}^\dagger(\varepsilon, \xi^\dagger)}{\partial \varepsilon}$$

$$\frac{\partial \mathcal{D}^\dagger(\dot{\xi^\dagger})}{\partial \dot{\xi^\dagger}} = -\frac{\partial \mathcal{W}^\dagger(\varepsilon, \xi^\dagger)}{\partial \xi^\dagger}, \quad \xi^\dagger(0) = 0. \qquad \text{(Solve using Backward Euler)}$$

Example

$\mathcal{W}^\dagger(\varepsilon, \xi^\dagger) = \frac{1}{2}(\varepsilon - \xi^\dagger)^2 + \frac{1}{2}(\xi^\dagger)^2; \quad \mathcal{D}^\dagger(\dot{\xi^\dagger}) = \frac{1}{2}(\dot{\xi^\dagger})^2$

$\sigma^\dagger = \varepsilon - \xi^\dagger$

$\dot{\xi^\dagger} = \varepsilon - 2\xi^\dagger; \quad \xi^\dagger(0) = 0$

A Sample from the 1D-Dataset

$\varepsilon(t) \in \mathbb{R}; \quad \sigma^{\dagger}(t) \in \mathbb{R}; \quad \xi^{\dagger}(t) \in \mathbb{R}$

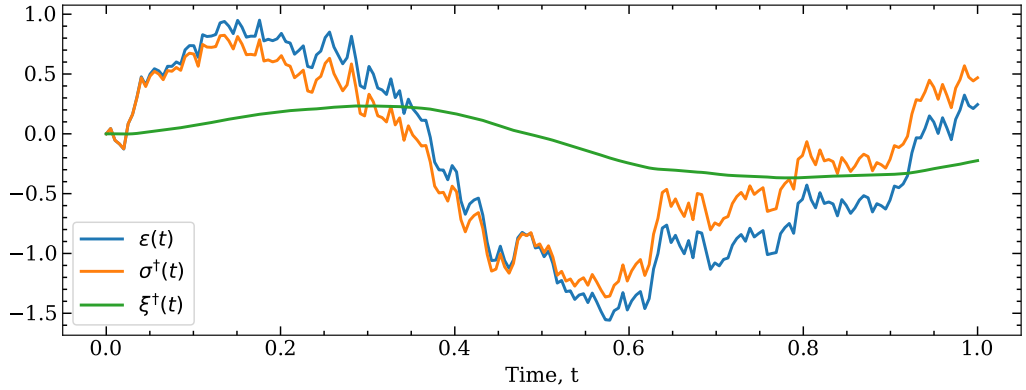


Figure: A trajectory sample from the dataset

Neural Network Training Process

- Dataset: $(\varepsilon_i, \sigma_i^\dagger)$ i = 1, . . . , N
- Prediction step (Forward Euler)

$$\frac{\partial \mathcal{D}}{\partial \dot{\xi}}\left(\frac{\xi(t_{n+1}) - \xi(t_n)}{\Delta t}; \alpha\right) = -\frac{\partial \mathcal{W}}{\partial \xi}(\varepsilon(t_n), \xi(t_n); \theta); \quad \xi(t_0) = 0$$

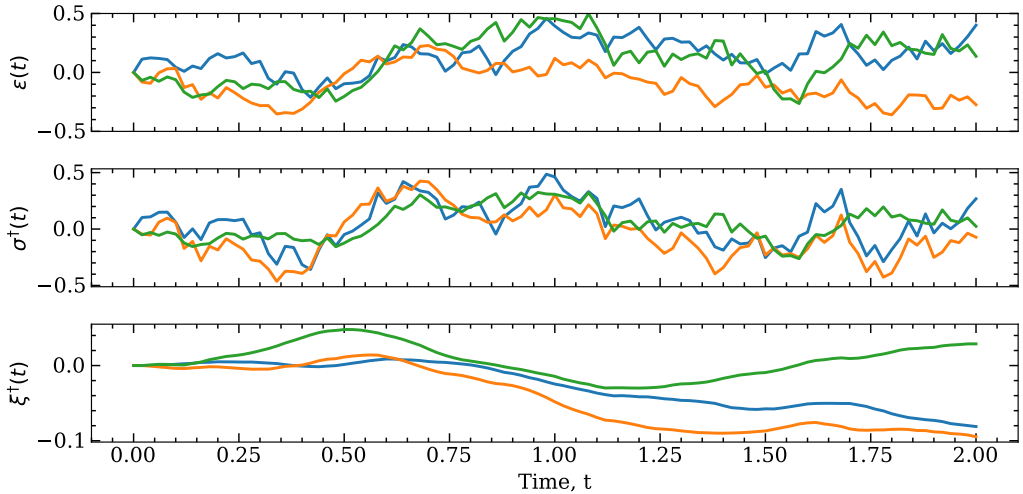
$$\sigma(t_n) = \frac{\partial \mathcal{W}}{\partial \varepsilon}(\varepsilon(t_n), \xi(t_n); \theta)$$

- Loss function: $\mathcal{L}(\theta, \alpha) = \frac{1}{N}\sum_{i=1}^{N} \|\sigma^\dagger - \sigma\|_{L^2(0,T)}$
- Neural Network Weight Update

$$\theta \leftarrow \theta - \frac{\partial \mathcal{L}}{\partial \theta}; \quad \beta \leftarrow \theta - \frac{\partial \mathcal{L}}{\partial \beta}$$

A Sample from the 2D-Dataset

$\varepsilon(t) \in \mathbb{R}^3, \quad \sigma^\dagger(t) \in \mathbb{R}^3, \quad \xi^\dagger(t) \in \mathbb{R}^3$



Visual Comparison of Stress

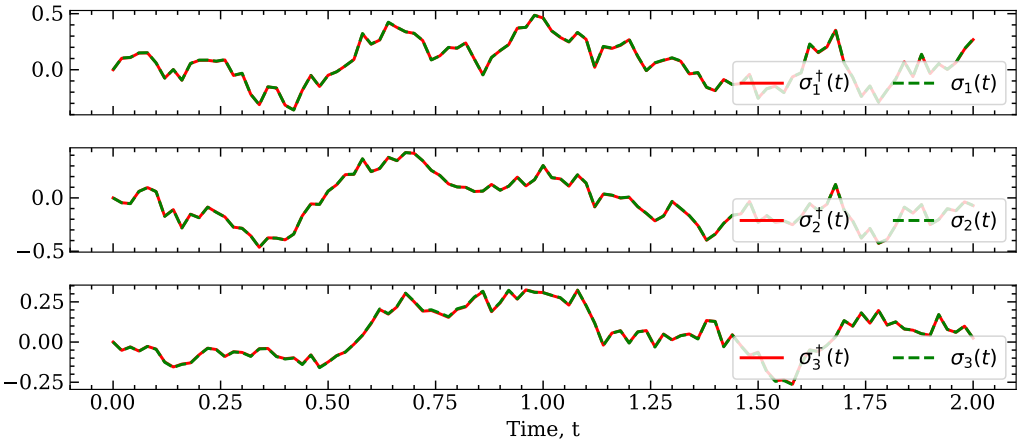


Figure: Comparison of true and predicted components of σ

Summary and Future Work

- Framework for learning a stable constitutive model of a non-Markovian process

Future work

- Learning homogenized response
- Enforce uniqueness of the internal variable
- Better understanding of quality of dataset and data generation

Challenges

- Slow training process